Tomato Yellow Leaf Curl Virus (TYLCV): Biology, Vector Interactions, Host Resistance and Emerging Management Strategies — A Review

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***Abstract*—Tomato yellow leaf curl virus (TYLCV; genus *Begomovirus*, family *Geminiviridae*) is among the most damaging pathogens of tomato (*Solanum lycopersicum*) worldwide. It is transmitted in a persistent, circulative manner by the *Bemisia tabaci* cryptic species complex and can cause severe yield loss in unprotected crops. This review summarizes recent work (2016–2026) on TYLCV genome function, vector transmission, global spread and recombination, host resistance (*Ty* genes), vector management, and biotechnological interventions including CRISPR/Cas genome editing and RNA interference. Existing studies are compared by approach, strengths and limitations, and the main research gaps are identified: durability of resistance under high disease pressure and heat, incomplete understanding of the virus–vector interface, insecticide resistance in whiteflies, and limited field validation of genome-edited and RNA-based protection. Durable control will require pyramiding of *Ty* loci, vector exclusion and field-tested editing strategies within one integrated management framework.**

***Index Terms*—TYLCV, begomovirus, *Bemisia tabaci*, Ty genes, resistance breakdown, CRISPR/Cas, RNA interference, integrated pest management.**

I. INTRODUCTION

Tomato (*Solanum lycopersicum*) is among the most widely grown vegetable crops, and its production is constrained by a broad range of pathogens. Among them, tomato yellow leaf curl disease (TYLCD) is consistently ranked as one of the most important, with the monopartite begomovirus *Tomato yellow leaf curl virus* (TYLCV) as its best-known causal agent [1], [2]. Typical symptoms are upward leaf curling, marginal chlorosis, severe stunting and flower drop; infection before flowering can cause near-total loss of marketable fruit, and reported losses in unprotected fields can approach complete crop failure.

TYLCV was first associated with disease in the Middle East and has since spread to tropical, subtropical and warm-temperate production areas on every inhabited continent [3], [4]. Its spread has been driven by trade in plant material, by the global movement of invasive *Bemisia tabaci* lineages, and by the virus's capacity to recombine with local begomoviruses [3], [5]. Control remains difficult because no curative treatment exists for infected plants, because whitefly vectors can transmit the virus after short feeding periods, and because resistance genes deployed in cultivars are not always durable [1], [6].

This review (i) summarizes TYLCV taxonomy, genome and vector biology, (ii) surveys recent literature on epidemiology, host resistance, vector management and biotechnology, (iii) compares the main control options, and (iv) identifies research gaps, challenges and priorities. The literature considered covers approximately the last ten years (2016–2026).

A. Scope and Review Approach

Literature was gathered by searching bibliographic and open-access sources for studies on TYLCV, begomovirus–whitefly interactions, tomato resistance genetics, vector management and genome-editing or RNA-based protection, with emphasis on work published from 2016 onward. Sources were grouped by theme and compared qualitatively because the studies differ widely in design, virus strains, tomato genotypes and environments, which precludes formal meta-analysis.

B. Symptoms and Economic Importance

Infected plants show leaflets that curl upward and inward into a cup shape, yellowing of leaf margins, shortened internodes, a bushy dwarfed habit, and reduced flower and fruit set. Symptom severity depends on plant age at infection: early infection causes the most severe stunting and the largest yield loss, whereas late infection mainly reduces fruit size and quality [1]. Because symptoms can resemble those caused by other begomoviruses, nutrient disorders or herbicide injury, molecular diagnosis is needed for reliable identification, and mixed infections with related viruses can generate recombinants [5].

II. BACKGROUND AND RELATED WORK

A. Taxonomy and Genome Organization

TYLCV belongs to the genus *Begomovirus*, family *Geminiviridae*, the largest family of plant viruses with circular single-stranded DNA genomes [7]. The family has recently been reorganized with the establishment of new genera [8], and begomoviruses are widely discussed as a highly adaptable and successful virus group [5]. TYLCV virions are twinned (geminate) particles formed by two incomplete icosahedra of about 20 × 30 nm that enclose a circular single-stranded DNA genome of roughly 2.8 kb. The genome carries six overlapping open reading frames (ORFs) on the virion-sense and complementary-sense strands, separated by an intergenic region that contains the origin of replication (Table I) [1].

TABLE I
TYLCV OPEN READING FRAMES AND FUNCTIONS

ORF	Protein	Main function
V1	Coat protein (CP)	Encapsidation, systemic movement, vector interaction
V2	Pre-coat protein	Movement; strong silencing suppressor
C1	Rep	Rolling-circle replication initiation
C2	TrAP	Transcription control; silencing suppression
C3	REn	Enhances replication
C4	Pathogenicity factor	Symptom induction, movement, silencing suppression

Because the virus replicates through a rolling-circle mechanism and relies on host DNA-replication machinery, the interaction with host factors is central to pathogenesis. Silencing suppression by V2, C2 and C4 counters host RNA-based defenses, and host chaperones such as heat shock proteins contribute to the establishment of infection [9]. High mutation and recombination rates allow TYLCV to adapt, and recombinants between TYLCV and related viruses are a recurring feature of its evolution [3], [5].

B. Host Range and Transmission Routes

Besides tomato, TYLCV infects other solanaceous crops (pepper, eggplant, tobacco), common bean, and weeds that serve as inoculum reservoirs. Natural spread is overwhelmingly by whiteflies, but TYLCV has also been shown to be seed-transmissible in tomato, which raises phytosanitary questions for seed production and trade [10].

C. The Whitefly Vector

TYLCV is transmitted by members of the *B. tabaci* cryptic species complex, notably MEAM1 and MED, in a persistent, circulative manner [11]. Virions are acquired from the phloem, pass through the stylet and foregut, cross the midgut epithelium into the hemolymph, and reach the salivary glands, from which they are inoculated into healthy plants (Fig. 1) [12]. A GroEL chaperone produced by whitefly endosymbionts is thought to protect virions in the hemolymph, and the efficiency of transmission differs between whitefly species [12], [11]. Availability of the MEAM1 genome has provided a resource for studying virus transmission, host adaptation and insecticide resistance [13].

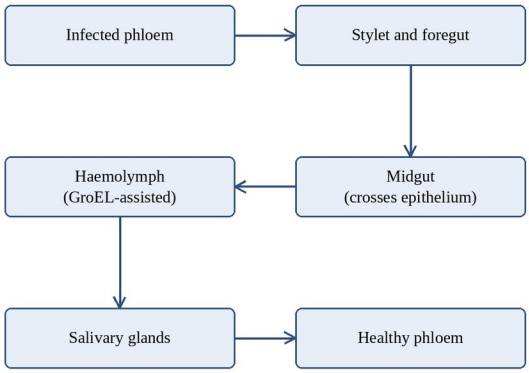


Fig. 1. Circulative transmission pathway of TYLCV in *Bemisia tabaci*.

Whitefly–plant interactions also influence epidemiology. Whitefly saliva carries effector proteins, and some of these suppress plant defenses; for example, a salivary effector (Bt56) promotes whitefly feeding by activating salicylic-acid signaling [14], and whiteflies can acquire a plant gene that neutralizes phenolic glucosides, helping them tolerate host chemical defenses [15]. Reviews of whitefly–plant interactions emphasize that such manipulation of host signaling can indirectly benefit virus spread [16].

III. LITERATURE REVIEW

A. Global Status and Epidemiology

Global reviews emphasize that TYLCD now involves multiple virus species and strains and threatens tomato production on all continents [1], [2]. Phylogeographic work indicates a Middle Eastern origin followed by repeated long-distance dispersal [3]. A genome-scale analysis of more than 800 TYLCV genomes reported that the Israel (IL) and Mild (Mld) strains are present across Africa, the Americas, Asia, Oceania and Europe, whereas recombinant strains were found mainly in the Middle East [4]. The related begomovirus complex *Tomato leaf curl New Delhi virus* illustrates how a different virus lineage can emerge as a threat to vegetable crops, with its own recombination and host-range dynamics [17].

B. Genetic Resistance and the Ty Loci

Resistance breeding has relied on loci introgressed from wild tomato relatives, designated *Ty-1* to *Ty-6* (Table II) [6], [18]. *Ty-1* and *Ty-3* are allelic and encode an RNA-dependent RNA polymerase that enhances methylation of viral DNA, giving broad geminivirus resistance [18]. *Ty-2*

encodes the NB-LRR protein TYNBS1 and confers resistance mediated by an immune-receptor mechanism [19]. *ty-5* is recessive and results from loss of function of the mRNA-surveillance factor Pelota (*SlPelo*) [20]. *Ty-6*, on chromosome 10, gives resistance to TYLCV and *Tomato mottle virus* [21] and was more recently attributed to a variant of DNA polymerase delta subunit 1 (SlPOLD1) [22].

TABLE II
MAJOR TYLCV RESISTANCE LOCI IN TOMATO

Locus	Source	Chr.	Mechanism / gene
Ty-1/Ty-3	*S. chilense*	6	RNA-dependent RNA polymerase; enhanced DNA methylation [18]
Ty-2	*S. habrochaites*	11	NB-LRR protein TYNBS1 [19]
ty-5	*S. peruvianum*	4	Pelota (*SlPelo*), recessive [20]
Ty-4	*S. chilense*	3	Minor-effect locus [6]
Ty-6	*S. chilense*	10	SlPOLD1 variant [21], [22]

Resistance is not absolute. Virus can still accumulate in plants carrying single *Ty* loci, and breakdown of *Ty-1*-mediated resistance has been reported under high inoculum pressure and in the presence of recombinant viruses [18], [6]. Resistance reduces disease severity more reliably than it blocks infection, which in turn allows low-level virus reservoirs to persist [6]. Pyramiding loci with different mechanisms is therefore the main strategy to improve durability [21], [18].

C. Vector Management

Whitefly control combines exclusion with fine-mesh nets, reflective mulches, trapping of adults, removal of weeds and infected plants, and insecticides. Insecticide resistance is a major constraint: *B. tabaci* has evolved resistance to multiple insecticide classes, and resistance-management programs are needed to preserve the value of remaining chemistries [23]. Recent surveys of whitefly populations on vegetable crops report resistance-associated mutations across several chemical classes, underscoring the need for integrated approaches [24]. Climate change is expected to alter insect pest distributions and population dynamics, which could change whitefly pressure and virus incidence in production areas [25].

D. Biotechnological Approaches

CRISPR/Cas. Two strategies have been explored. In the first, Cas9 is programmed to cleave the viral genome. Targeting TYLCV sequences in tomato and *Nicotiana benthamiana* reduced viral accumulation and symptoms, with interference reported to persist over generations, although targeting coding regions risks generating escape variants [26], [27]. In the second, host susceptibility genes are edited; knocking out *SlPelo* in tomato restricted viral DNA accumulation and conferred TYLCV resistance [20]. Because TYLCV is a DNA virus, RNA-targeting systems such as Cas13a, demonstrated against RNA viruses [28], would act on viral transcripts rather than the replicating genome.

RNA interference. Transgenic hairpin constructs against viral genes can confer strong resistance, but topical dsRNA approaches have given inconsistent results: one comparison reported no significant reduction of TYLCV-IL accumulation after external dsRNA, whereas a hairpin construct against *Rep* gave high resistance [29]. Clay-nanosheet delivery of dsRNA has shown sustained protection against other plant viruses and is a candidate delivery platform for TYLCV [30].

E. Resistance Screening and Breeding Pipelines

Developing TYLCV-resistant cultivars involves several stages: identification of resistance sources in wild relatives, introgression into elite lines, phenotyping under controlled inoculation, and stacking of loci. Germplasm surveys have identified additional resistance sources beyond the six named loci, which broadens the pool available for pyramiding [6]. Molecular markers linked to *Ty-1/Ty-3*, *Ty-2*, *ty-5* and *Ty-6* allow early selection and reduce the number of generations needed to combine loci [18], [21]. Because resistance can be influenced by temperature and by the infecting strain, phenotyping across environments and with diverse isolates is essential before a stacked combination is released [18], [4].

Introgression from wild species also carries linkage drag, in which undesirable alleles travel with the resistance gene. Cloning of *Ty-2* and *Ty-6* provides the sequence information to design precise markers and, in principle, to deploy the genes through transformation or editing without linked material [19], [22]. Editing of a susceptibility gene such as *SlPelo* offers a complementary route that bypasses introgression altogether, although the agronomic consequences of the edit must be evaluated [20].

F. Host Factors and Molecular Interactions

Infection depends on the interplay between viral proteins and host pathways. V2 and the other suppressors interfere with RNA-silencing and with epigenetic defenses, and the resistance conferred by *Ty-1/Ty-3* acts by strengthening exactly this line of defense through increased DNA methylation [18]. Host chaperones and heat shock proteins modulate viral accumulation [9], and the susceptibility gene Pelota is required for efficient infection, which is why its loss is protective [20]. Such host dependencies are attractive targets because they are less easily overcome by viral

mutation than virus-specific sequences, although loss of host functions may carry fitness costs [20], [27].

G. Diagnostics and Surveillance

Reliable detection underpins both breeding and epidemiological monitoring. PCR-based assays, quantitative PCR and sequencing of the intergenic region and coat-protein gene are routinely used to identify TYLCV and to distinguish strains and recombinants [4], [8]. Genome-scale sequencing has made it possible to reconstruct dispersal routes and detect recombination events, and this approach is the basis of current surveillance proposals [4], [3]. The detection of seed-borne virus further motivates testing of seed lots in addition to field monitoring [10].

H. Integrated Management in Practice

Field recommendations combine several layers: use of virus-free transplants raised under insect-proof nets, planting of resistant cultivars, reflective mulches during early growth, rogueing of infected plants, weed control around fields, and a crop-free period to break the virus cycle. No single layer is sufficient. Resistant cultivars lower the proportion of infected plants but may still support virus replication, so vector suppression and sanitation remain necessary to limit the build-up of inoculum [6], [1]. Vector suppression relies on rotating insecticides with different modes of action to slow resistance selection [23], which is consistent with surveillance data showing resistance to several chemical classes [24].

IV. COMPARATIVE ANALYSIS OF EXISTING RESEARCH

Table III compares the main control strategies discussed above in terms of evidence, strengths and limitations. Host resistance is the most mature and economical tool, but it is vulnerable to recombinants and environmental effects. Vector control is effective in the short term yet depends on chemistries under resistance pressure and on infrastructure that many growers cannot afford. Genome editing and RNA-based protection are promising but have mostly been evaluated under controlled conditions.

TABLE III
COMPARISON OF TYLCV MANAGEMENT STRATEGIES

Strategy	Strengths	Limitations
Ty-1/Ty-3 [18]	Broad geminivirus resistance; widely deployed	Breakdown under high pressure and recombinants
Ty-2 [19]	Defined NB-LRR mechanism	Narrower virus spectrum
ty-5/*Ty-6* [20], [21]	New mechanisms; effective when stacked	Few multi-environment durability data
CRISPR, host gene [20]	Non-transgenic potential; restricts viral DNA	Regulation; pleiotropy of edits
CRISPR, viral target [26]	Heritable interference	Escape variants; delivery
Topical dsRNA [29], [30]	Non-transgenic, rapid	Inconsistent protection; cost
Insecticides [23], [24]	Rapid vector suppression	Resistance; environmental load
Exclusion/mulches [12]	No chemical input	Cost; ventilation; short window

Three cross-cutting observations emerge from the comparison. First, strategies that act at different points of the disease cycle are complementary: host resistance limits virus accumulation in the plant, vector control limits acquisition and inoculation, and sanitation reduces inoculum sources. Second, evidence quality is uneven: resistance genetics is supported by mapped and cloned loci [19], [21], whereas topical RNA approaches rest on a small number of small-scale studies [29], [30]. Third, many evaluations are strain- and environment-specific, so a locus that performs well against one TYLCV strain or at one temperature cannot be assumed to be durable elsewhere [18], [4].

The management of the related *Tomato leaf curl New Delhi virus* shows that lessons from TYLCV, including vector exclusion, resistance pyramiding and recombination monitoring, are transferable to other emerging begomoviruses [17], [5]. Conversely, the finding that begomovirus transmission depends on the vector species [11] implies that control measures tuned to MEAM1 may not be equally effective against MED or other members of the complex.

V. RESEARCH GAPS

1) *Durability of resistance.* Few studies evaluate stacked *Ty* loci across seasons, temperatures and virus populations, although recombinants are known to overcome single loci [18], [6].

2) *Virus–vector interface.* Whitefly effectors, detoxification genes and symbiont proteins have been characterized, but their specific roles in TYLCV acquisition and transmission are not fully resolved [14], [15], [12].

3) *Seed and non-vector routes.* The epidemiological importance of seed transmission is poorly quantified [10].

4) *Climate effects.* Temperature and climate change are expected to modify vector dynamics [25], but multi-year field data linking climate to TYLCV incidence and resistance performance are scarce.

5) *Field validation.* Most CRISPR and RNA-based protection has been tested in controlled environments [20], [26], [29].

6) *Surveillance.* Recombinant detection is geographically uneven [4], [3].

VI. CHALLENGES AND LIMITATIONS

Resistance is partial and viral replication is not fully blocked, allowing selection of adapted variants [18]. Insecticide resistance erodes chemical options [23], [24]. Regulatory and public-acceptance barriers affect genome-edited cultivars, and the cost of net houses limits their use by smallholders. Research is concentrated on a few strains (IL, Mld), leaving other regions and recombinants poorly characterized [4]. Finally, the heterogeneity of experimental conditions makes direct comparison between studies difficult, and this review is limited by reliance on published summaries for some sources.

VII. FUTURE RESEARCH DIRECTIONS

Priorities include: (i) pyramiding *Ty-1/Ty-3*, *Ty-2*, *ty-5* and *Ty-6* by marker-assisted selection, with multi-location and multi-temperature durability testing [21], [19], [22]; (ii) editing *SlPelo* and multiplexed viral targets with field trials and off-target assessment [20], [27]; (iii) nanocarrier-based dsRNA delivery against virus and vector [30]; (iv) functional studies of whitefly effectors and endosymbiont proteins as targets for blocking transmission [14], [15], [12]; (v) genomic surveillance for recombinants and early-warning systems [4], [3]; and (vi) integration of resistant cultivars, exclusion and biological control with insecticide-resistance management [23].

An additional priority is the economic and social side of management. Most evaluation to date is agronomic; cost–benefit comparisons of resistant cultivars, net houses and chemical programs for smallholder versus commercial production are scarce, although these determine which options growers can adopt. Training of growers in monitoring, early removal of infected plants and rational insecticide use should accompany technical advances so that gains from new resistance genes and editing tools are not lost to premature breakdown [23], [1].

TABLE IV
PROPOSED RESEARCH AGENDA

Horizon	Priority	Key evidence
Short term	Multi-site durability trials of stacked *Ty* lines; recombinant surveillance	[18], [4]
Short term	Rotation-based insecticide resistance management	[23], [24]
Medium term	Field testing of edited *SlPelo* and other host-factor edits	[20]
Medium term	Functional validation of whitefly effectors and symbiont proteins	[14], [15], [12]
Long term	Nanocarrier dsRNA against virus and vector	[30], [29]
Long term	Climate-informed epidemiological models	[25]

VIII. CONCLUSION

TYLCV remains a major constraint on tomato production because of efficient whitefly transmission, rapid evolution and recombination, and the incomplete durability of single resistance sources. New resistance mechanisms (Pelota, POLD1), genome-editing demonstrations and improved knowledge of whitefly–plant interactions provide a stronger toolkit, but field-validated, multi-layered strategies are still needed. Stacked host resistance, vector exclusion and biological control, supported by genomic surveillance, offer the most realistic route to durable protection.

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